



Week 3 – Classification & Model Evaluation

Learning Objectives



- Understand classification concepts
- Training vs Testing datasets
- Evaluation metrics
- Confusion matrix

Introduction to Classification

- ▶ Supervised learning technique
- ▶ Predicts categorical outputs
- ▶ Examples: Spam detection, Disease prediction

Types of Classification

- ▶ Binary classification
- ▶ Multiclass classification
- ▶ Multilabel classification

Training vs Testing Dataset

- ▶ Training data: learn patterns
- ▶ Testing data: evaluate performance
- ▶ Typical split: 70/30

Importance of Data Splitting

- ▶ Prevents overfitting
- ▶ Ensures generalization
- ▶ Improves real-world performance

Confusion Matrix

- ▶ TP, TN, FP, FN
- ▶ Used to evaluate classification models

Evaluation Metrics

- ▶ Accuracy
- ▶ Precision
- ▶ Recall
- ▶ F1-score

Accuracy

▶ $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$

Precision & Recall

- ▶ Precision = $TP / (TP + FP)$
- ▶ Recall = $TP / (TP + FN)$

F1 Score

►
$$F1 = 2 * (Precision * Recall) / (Precision + Recall)$$

Activities

- ▶ Confusion matrix exercise
- ▶ Train simple classifier
- ▶ Evaluate performance